

Investigation of coupling efficiency in laser beam welding of copper materials using brilliant infrared and green laser radiation

Florian Kaufmann^{a,b,*}, Robert Möttingdörfer^c, Stephan Roth^{a,d}, Michael Schmidt^{a,c,d}

^a Bayerisches Laserzentrum GmbH (blz), Konrad-Zuse-Str. 2-6, 91052 Erlangen, Germany

^b Friedrich-Alexander Universität Erlangen-Nürnberg (FAU), Schloßplatz 4, 91054 Erlangen, Germany

^c Institute of Photonic Technologies (LPT), Friedrich-Alexander Universität Erlangen-Nürnberg, Konrad-Zuse-Str. 3/5, 91052 Erlangen, Germany

^d Erlangen Graduate School in Advanced Optical Technologies (SAOT), Paul-Gordan-Str. 6, 91052 Erlangen, Germany

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ABSTRACT

One of the fundamental effects of laser material processing is laser beam absorption. The absorptivity in the laser welding process is subjected to temporal and spatial fluctuations due to temperature and phase changes and is furthermore dependent on the wavelength used. Currently, experimental absorptivity data for copper at high temperatures is rarely available, and discussions of investigation are largely based on theoretical assumptions. However, this property is essential for a fundamental understanding of the phenomena involved in laser-matter interaction and for efficient process design, especially for the emerging manufacturing processes for electric vehicles, where copper materials are widely used.

In this work, a specially designed, low-cost experimental arrangement was built to measure the laser absorption during the process in different welding modes, which enables the derivation of coupling efficiency values in these regimes. Brilliant laser beam sources with $\lambda = 515$ nm and $\lambda = 1030$ nm radiation and comparable spot characteristics on the workpiece were used for the investigation. Results captured with microsecond resolution show several important features, including the points of melting and vapor capillary formation for both pulsed and continuous process guiding. General tendencies known from literature could be confirmed by our measurements, and the comparison of the absorptivity data with microscopic inspection and a weld seam cross-sectional analysis shows a significant correlation of the characteristics.

1. Introduction

Global transportation CO₂ emissions have increased by >70 % since 1990, accounting for >20 % of global CO₂ emissions from burning fossil fuels in 2022. Owing to the fact that the fuel consumption is correlated with the amount of CO₂ and other greenhouse gas emissions, which are directly linked to the climate change phenomenon, the automotive sector began to switch the focus from traditional internal combustion engines to alternative propulsion systems [1]. Sales of these electrically powered vehicles are steadily increasing in Europe and worldwide. By 2030, every second new vehicle worldwide is foreseen to be electrically powered [2]. This trend is leading to a significant socio-economical change, since the electrification of powertrains fundamentally affects the types of materials, their use and manufacturing processes, and enables new vehicle concepts [3].

Ever-increasing demands on these vehicles in terms of weight

reduction, range extension and additional features as for example fast-charging capability imply shortened development cycles, making highly flexible, reliable, and automated production machinery an essential tool for project realization [4]. From this perspective, laser-based manufacturing is a key technology, establishing this process family as the state of the art in the production of electric vehicles. Laser beam sources are used in various applications such as hairpin welding [5], cutting and structuring of lithium-ion electrodes [6], and battery busbar welding [7] for achieving sophisticated goals while ensuring high quality standards concerning electrical and mechanical requirements.

In laser beam welding, the extensive use of copper materials in particular poses new challenges for process design. The variety of connection types for mobility ranges from bare copper (Cu-ETP/ Cu-OF) to tin-plated surfaces, alloys like bronze, and mixed joints with aluminum or steel in thicknesses from a few micrometers to several millimeters, which need to be joined. To weld these materials with

* Corresponding author at: Bayerisches Laserzentrum GmbH (blz), Konrad-Zuse-Str. 2-6, 91052 Erlangen, Germany.

E-mail address: florian.1.kaufmann@fau.de (F. Kaufmann).

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shortest cycle times and minimized energy input, laser beam sources with highest beam quality in the near infrared wavelength range (1030–1070 nm) are widely used [8]. In addition, visible wavelength lasers have recently been made available to the industry in green [9] and blue wavelengths [10], targeted at the processing of highly reflective metals.

Despite the various options in laser welding technology, fundamental studies investigating the comparison of process characteristics at different wavelengths using equal boundary conditions and efficiency considerations are limited in the literature. One of the basic properties of laser welding is the absolute value of the absorption of irradiated energy, which involves complex light-matter interaction with changing aggregate states and can include vapor and particle interactions, making it difficult to quantify. Nevertheless, knowledge of this value is of vital importance for optimizing process parameters, ensuring efficient energy transfer, and achieving high-quality welds.

The present work aims to assess in situ absorptivity data during copper welding using brilliant beam sources thus contributing to a deeper understanding of the process. For this purpose, a combined welding and measuring setup based on integrating spheres was designed and calibrated. The time-resolved absorptivity data applying different process guiding with infrared and green laser radiation are discussed highlighting important process features and benefits of the systems.

2. State of the art

2.1. Laser beam welding of copper and influence of the wavelength

The laser beam welding process belongs to the group of fusion welding. In general, the process regime in laser welding is strongly sensitive to the applied process parameters, which is challenging for process development, as the prevailing process regime strongly influences the process result [11]. There is a division into two categories: heat conduction welding (HCW), also known as conduction mode, and deep penetration welding (DPW), also known as keyhole mode. The laser welding processes can basically be designed in two different operating modes. A distinction is made between continuous wave (cw) operation and pulsed mode. In continuous wave mode, laser radiation is continuously applied to the sample during the application of a feed rate. In pulsed mode, the laser emission is interrupted at regular intervals. This can be used for single points or quasi-continuous lines to create interfaces similar to the former mode [12]. The latter operation mode helps to reduce the energy input, which may be advantageous for heat- or crack-sensitive joints, for example, and to control dynamics.

Copper materials pose a particular challenge for ensuring a stable laser beam welding process. The challenges are attributed to its physical properties. The high thermal conductivity and low melt viscosity of copper lead to rapid heat dissipation and low suppression of process dynamics, especially during deep penetration welding. Therefore, spatter formation and melt ejection are frequently observed defects that particularly occur at moderate feed rates ($v \leq 6$ m/min) to an excessive extent [13]. In addition, a lack of reproducibility is reported when using conventional NIR lasers, which are well researched for material processing. Here, the optical properties of copper impede the welding process [14].

In order to achieve an efficient welding process, the irradiated laser energy must be introduced into the sample and converted into heat as effectively as possible. This conversion of laser energy into process heat is referred to as energy coupling and is predominantly determined by the power that is absorbed in the workpiece. A measure of the power available for the welding process (P_{Abs}) is the absorptance or absorptivity A . It is defined as the ratio between the power deposited in the workpiece and the power of the incident radiation and is calculated according to Eq. (1)

$$A = P_{Abs}/P, \quad (1)$$

with A : absorptivity (–), P_{Abs} : absorbed laser power in W, and P : incident power in W.

Since the thermal energy introduced has an additional effect on the temperature-dependent optical material properties, the phase transitions caused, and the associated geometrical properties, there is a dynamic interaction between the absorptivity and the temperature profile generated on the metal surface during the laser beam welding [15]. The absorptivity A can assume any value between 0 and 1. The remaining laser power is either reflected at the material surface or transmitted through it, as described by Eq. (2) [16].

$$1 = A + R + T, \quad (2)$$

with A : absorptivity (–), R : reflectivity (–), and T : transmissivity (–), where R and T have the same range of values as A . When welding metals with a thickness considerably larger than the wavelength used, the transmission is negligible, so the formula is shortened to the reflectivity and the absorptivity, with their addition equaling one.

The absorptivity depends, among other factors, on the material, the temperature, and the wavelength used. Fig. 1 shows the material specific absorptivity of various metals at room temperature in the wavelength range from 0.1 to 10 μm at perpendicular beam incidence. In addition, the diagram on the right shows the temperature-dependent values for pure copper for 0.5 μm and 1 μm wavelength.

In the wavelength range $\lambda \geq 1$ μm (typical NIR laser), the absorptivity of copper at room temperature is below 5 % and significantly lower than the value for steel. It exhibits an erratic increase at the transition from the solid to the liquid state to approx. 14 %, associated with an abrupt decrease in thermal conductivity in the melt [17]. This fact may negatively affect the laser welding process with unfavorable parameter setting due to an accumulation of the heat in the interaction zone [18]. In addition, these boundary conditions make it difficult to initiate and maintain a stable heat conduction welding process on copper.

The use of novel wavelengths for copper welding with high power beam sources is an emerging topic in literature. Due to recently available laser beam sources in the green (515 nm) [9] and blue (450 nm) [21] wavelength range, absorptivity levels of 40–60 % at ambient conditions are accessible for the process start. Current issues related to blue diode lasers are the low beam quality and the associated difficulties in launching into small process fibers with high laser powers [8]. Therefore, this work focuses on brilliant beam sources with green laser radiation. For 515 nm wavelength, there is also no change in absorptivity with an increase in temperature of the same order of magnitude as for NIR radiation. This circumstance has already been proven to enable a more precise and controllable copper welding process, and to achieve heat conduction welding mode for Cu-ETP with unsurpassed quality [22].

The absorptivity for 515 nm and 1030 nm during heat conduction welding of copper comes close to the literature values, as expected, since there is a single interaction between the laser beam and the sample material [23]. As the effect of multiple reflection in the vapor capillary comes into play during deep penetration welding, absorptivity values close to 1 can be achieved in the welding process [24]. The total absorbed proportion of the irradiated energy during a laser beam welding process is also referred to as the degree of energy coupling η_A (coupling efficiency). Although copper laser welding technology has reached a certain level of maturity, questions regarding an efficient process design, especially in direct comparison of brilliant infrared and green laser beam sources, have not yet been addressed, whereby the time dependent measurement of the property absorptivity is of great value.

2.2. Absorptivity measurement

The measurement of absorptivity is in general a complex task since it

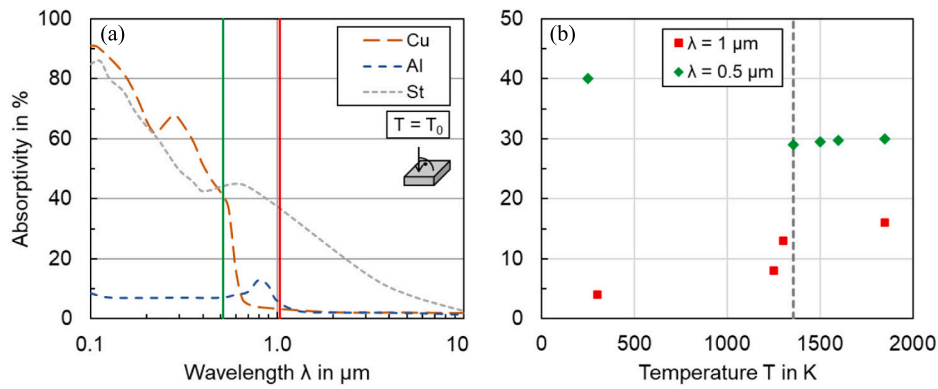


Fig. 1. (a) Absorptivity of various metals in dependence on the wavelength at room temperature and (b) temperature dependence of the absorptivity of copper at 0.5 μm and 1.0 μm [19,20].

depends on multiple factors. As already pointed out, surface roughness, its temperature, wavelength, light polarization, pressure conditions, state of aggregation, and the presence of oxide layers are among the characteristics [25]. Absorptivity measurements are typically divided into two main groups, direct and indirect methods. In direct methods, the energy introduced into the workpiece is measured. When using an indirect method, physical relationships with other radiative properties are taken into account. For example, the reflectivity or emissivity of the sample is measured, and the absorptivity is calculated therefrom.

The calorimetric (direct absorption) measurement is based on the determination of laser-induced sample heating. A sample is positioned in a measuring chamber and continuously irradiated with a laser for a certain time t , assuming the heat introduced is constant over time, and recorded using a thermocouple or pyrometer. Using the mass and heat capacity of the material, the absorbed energy can be calculated. Engler et al. used thermocouples attached to copper samples to measure the temperature increase due to laser irradiation and derived the energy coupling efficiency [23]. A nonlinear impact of energy coupling at varied feed rate on mild steel was observed using fast thermocouple measurements [26]. Disadvantages of thermocouple measurements are that a model considering heat conduction and dissipation effects is required, and that this principle is not suitable for measurements at high temperatures [27].

The measurement of emissivity is based on Kirchhoff's law, which states that for a closed system in thermal equilibrium, the amount of emitted energy is equal to the amount of absorbed energy. In general, the law relates the radiative properties of materials but cannot consider polarization effects [28]. By comparing the emitted radiation of the sample with a black body of equal temperature, the absorptivity can be determined. This principle requires long measuring times, as a thermal equilibrium must be maintained, and the implementation of high-precision temperature measurement is necessary [29]. For the radiometric (indirect absorption) measurements, the shortened version of Eq. (2) must apply. The transmission of the irradiated wavelength must be negligible, which is typically true for metals ($A = 1 - R$, see Section 2.1), allowing the absorptivity to be determined directly from the measurement of the reflectivity. Using an integrating sphere is a widely used measuring setup for reflectance measurement. A sample is placed and irradiated in the hollow sphere with sample and measurement ports, coated with a diffuse reflective layer on the inside, whereby the reflected portion of the irradiated light is diffusely scattered on the inside of the sphere. The radiation intensity of the integrating sphere is inversely proportional to its diameter so the smaller the sphere, the stronger the signal. The spectral reflectivity can be detected using photodiodes, for example, so after previous calibration the reflectivity can be determined from the measurement data [27]. Typical error due to radiation losses or light scattering sum up to 5–6 % [30].

Using an integrating sphere setup, the dynamic, absolute absorbed

power with sub-microsecond resolution during a 10 ms laser spot weld in stainless steel from conduction to keyhole welding condition was measured in [31], observing coupling efficiency increase from around 30 % to values approaching 90 %. To prevent coaxial back-reflected radiation during a welding process from escaping from the sphere via the inlet port, a beam splitter is integrated to redirect possible exiting reflections from the welding sphere into a secondary integrating sphere. This setup enables the measurement of diffuse and direct reflectivity. To protect the gold coating of the spheres from a continuous buildup of metal vapor, absorbing reflections and yielding an artificially high level of weld energy absorption, a cylinder of fused silica was placed over the sample. Overall energy transfer efficiencies measured by this method for spot welds (pulsed mode of operation) show good agreement with calorimetric and thermal expansion measurements [32]. The identified parameter dependencies are in accordance with the results reported in [33], extrapolating the absorptivity based on the evaluation of cross-sections and a thermal simulation. Wittemer et al. investigated absorptivity measurement with a similar integrating sphere setup using the combination with the back-reflection sensor of the NIR fiber laser used for welding for in situ process regime identification. The spatial mapping of the integrating sphere signal to melt tracks of AISi316L facilitated the targeted analysis of correlations between characteristic signal peaks and visible surface irregularities in the melt track [34]. Helm studied the influence of laser-based pre-processing and subsequent welding with a commercial fiber laser and an ns-pulsed laser source on copper using an integrating sphere setup. It was observed that the initial back reflection can be completely eliminated by surface structuring, and the degree of energy coupling can be significantly increased in dependence on the configuration of pre-processing and welding setup [35]. Nordet et al. [36] performed absorptivity measurements during laser powder bed fusion of pure copper with a 1 kW cw green laser, comparing the results with bead on plate welds. The authors observed a significant increase in absorptivity evolution with temperature in different welding modes, especially in the HCW regime in the copper powder bed compared to bare substrate.

In summary, the above-mentioned studies show that the use of an integrating sphere measurement setup is a viable solution for the in situ measurement of absorptivity during a welding process, however, the experimental effort is high and occurring weld irregularities (spatter, melt ejection) may damage the sensitive components. There is a knowledge gap in literature both about the absolute values of the absorptivity under actual processing conditions, and the predominant physical mechanisms involved in laser beam welding of copper using high power infrared and emerging green wavelength lasers. Measurements of the absorptivity above the melting temperature of metals are rarely performed, but are necessary for a better understanding of the process, simulation, and process optimization [37]. Thus, the objective of this work is the experimental investigation of absorptivity and

coupling efficiency in direct comparison of different laser wavelengths in copper welding using a specially developed, low-cost experimental setup based on integrating spheres and the correlation to metallographic analyses of the weld seams. The welding setup is capable of applying brilliant laser beam sources with $\lambda = 515$ nm and $\lambda = 1030$ nm radiation and comparable spot characteristics onto the workpiece. Measurements were captured with microsecond resolution in a selected parameter range covering important features like melting and vapor capillary formation in both pulsed and continuous mode operation.

3. Experimental setup and methods

3.1. Laser beam welding setup

In the experimental setup, a disc laser TruDisk 6001 and a frequency-doubled disk laser TruDisk 3022 (Trumpf GmbH, Ditzingen, Germany) with a maximum cw-output power of 6 kW at $\lambda = 1030$ nm and 3 kW at $\lambda = 515$ nm wavelength respectively were used as radiation sources. The system properties and properties of the optical setups for laser beam welding are presented in Table 1.

For the beam deflection, two galvanometer scanner systems were used. The beam-delivery fibers coupled to the scanning optics (Trumpf PFO 33–2) had a core diameter of $d_{LLK} = 150$ μ m. The f-Theta lenses had a focal length of $f_F = 255$ mm, resulting in a magnification ratio of 1:1.7 in combination with the collimation focal length of $f_C = 150$ mm each. Given that the fiber end is imaged onto the focal plane, a spot diameter of $d_F = 257.1$ μ m for the green laser and $d_F = 255.8$ μ m for the infrared laser respectively in top hat intensity distribution were measured for $P_L = 1000$ W laser power using a Primes Focus Monitor FM+ (Primes GmbH, Pfungstadt, Germany). This provides equal boundary conditions for the laser beam welding investigations.

3.2. Integrating sphere measurement setup

The in-house developed test rig used consists of two integrating spheres made from additively manufactured PLA (BASF Ultrafuse black, BASF SE, Ludwigshafen, Germany) halves with an inner diameter of $d_{Sphere} = 100$ mm and customized port arrangement. The diameters are $d_{In} = 25$ mm for the inlet ports and $d_{Out} = 12$ mm for the sensing ports. The inside of these spheres was prepared using an adhesion promoter and several successive layers of gap filler to obtain a smooth, sphere-like surface. A BaSO₄ paint [38] was applied on top, resulting in a uniform, diffuse surface after the application of several thin layers (approx. 20). The final surface, illuminated with a probe laser beam with a wavelength of 532 nm, is shown in Fig. 2.

A baffle, coated using the same procedure as described above, was installed inside the lower sphere to protect the detection port from directly scattered light and the particle plume. A schematic sketch of the complete measurement setup including welding optics is shown in Fig. 3. The sample mount is designed to be removable, with the inner surface (rim) also having a barium sulphate (BaSO₄) coating to create a Lambertian surface.

In the experimental procedure, the welding sample was placed in the sample mount, which then was attached to the lower integrating sphere from the bottom. The focal plane was positioned on the specimen's

upper surface during all experiments. The processing laser beam first hits a beam splitter, mounted at an angle of 45° to the material surface being processed. This beam splitter was adapted according to the laser wavelength used ($\lambda = 515$ nm: BSN10; $\lambda = 1030$ nm: BSN11, Thorlabs GmbH, Newton, NJ, United States). In investigations using the green laser, the beam splitter had a transmission of 88,18 % and a reflectivity of 11,37 %, with the infrared laser the values were a transmission of 88,16 % and a reflectivity of 10.88 %, respectively. The transmitted power was introduced into the lower integrating sphere and subsequently into the workpiece, while the reflected power was dissipated using a heat sink (air-cooled power meter). In the following, all power values refer to this transmitted power (P) as this is the relevant parameter for the welding process.

During a welding process, the incident laser radiation is partially reflected and absorbed by the workpiece, as explained in Section 2.1. The portion of the off-normal scattered, reflected radiation dissipated to the sides is homogeneously distributed to an average intensity by multiple reflections within the lower sphere. This intensity was measured with the lower photodiode, which is referred to as diffusely reflected radiation. The coaxially reflected radiation is reflected back onto the beam splitter, where again ~ 88 % of the radiation is transmitted, while the remaining portion is reflected into the upper integrating sphere. Again, this radiation is diffusely reflected at the sphere surface to an average intensity, recorded with the upper photodiode, subsequently referred to as directly reflected radiation. The measurements were performed using two Si-photodiodes of the type DET100A (Thorlabs). The analog voltage signal output by the photodiodes, dependent on the radiation intensity inside the sphere, was recorded using a PicoScope 6404C digital oscilloscope from Pico Technology. Both photodiodes were equipped with appropriate bandpass filters for the laser wavelength (Thorlabs FBH515–10 for $\lambda = 515$ nm; Thorlabs FLH1030–10 for $\lambda = 1030$ nm, with a FWHM bandwidth of 10 nm each) for noise suppression. Moreover, optical filters of known density (Thorlabs NEK01, optical density OD 0.1 - OD 4) were used to reduce the photodiode signal in the case of signal overload.

To enable temporal and spatial mapping between the sensor values and the resulting weld seam during data processing, the analog output of the DLC board of the laser systems was used. Therefore, the data acquisition trigger is based on an output voltage that is proportional to the temporal progression of the laser power generated by the laser source. All signals were sampled with a frequency of $f_M = 10$ MHz by the oscilloscope.

The integrating sphere setup was flushed with nitrogen immediately after an experiment, which was supplied through an inlet in the beam splitter mount (not visible in Fig. 3). This procedure was implemented to remove welding emissions (particle plume) from the spheres' interior, avoiding contamination of its inner (BaSO₄) surface. The nitrogen flow was stopped when a new sample was prepared and inserted into the welding position.

For the investigations, copper samples of Cu-ETP and CuSn6 (Gemmet Metalle, Fürth, Germany) with a dimension of 15 mm $\times$ 15 mm, and a thickness of $t = 3$ mm were prepared from sheet material. Before laser welding, the samples were cleaned with isopropyl alcohol. A rectangular pulse shape with a pulse duration of 10 ms with varied pulse energy for both infrared and green laser radiation was used for the spot welds. Cw-welds were created keeping the sample fixed while the laser beam was moved at a constant feed rate v . Thus, partial penetration welds of $l_{Weld} = 5$ mm in length were generated using the different laser wavelengths with laser power ranging from $P = 0.5$ –6 kW (note: up to the maximum output power P_{max} of the laser systems, up to 3 kW for green laser radiation respectively), and feed rate varied between $v = 6$ –18 m/min. The processing parameter combinations were selected based on preliminary tests, including the transition from heat conduction to deep penetration welding, and avoiding a complete penetration of the copper specimen. This transition was additionally investigated with cw-welds at a constant feed rate and linearly increased laser power ($\Delta P = 1000$ W).

Table 1

Parameters and properties of the laser beam sources and the optical setups used within this work.

Parameter	Unit	Trumpf TruDisk 3022	Trumpf TruDisk 6001
Laser medium	–	Disc	Disc
Wavelength λ	nm	515	1030
Laser power P_{max}	W	3000	6000
Fiber diameter d_{LLK}	μ m	150	150
Focal diameter d_F	μ m	257.1	255.8

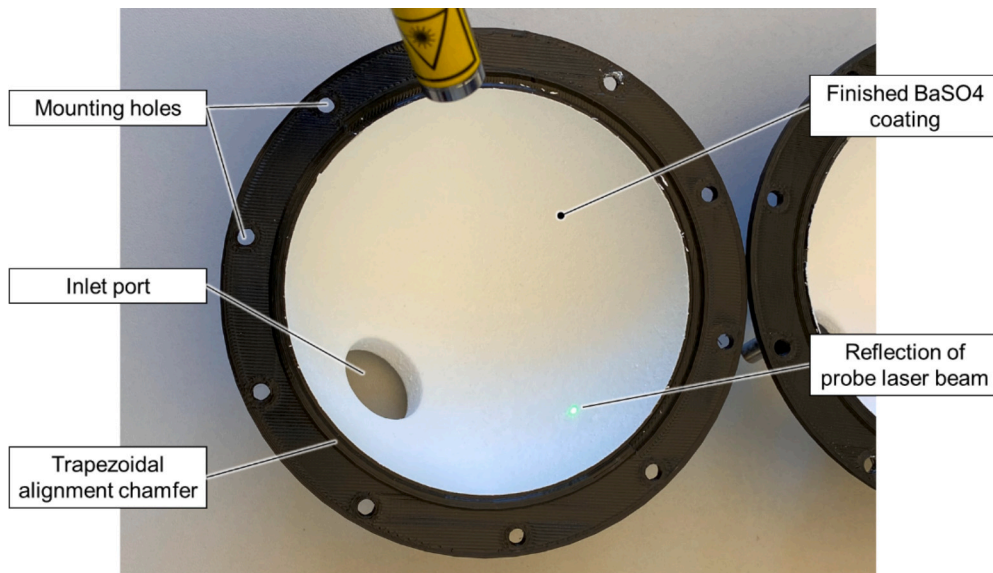


Fig. 2. Photograph of the interior of two halves of our custom-made integrating sphere setup with open inlet port and green probe laser.

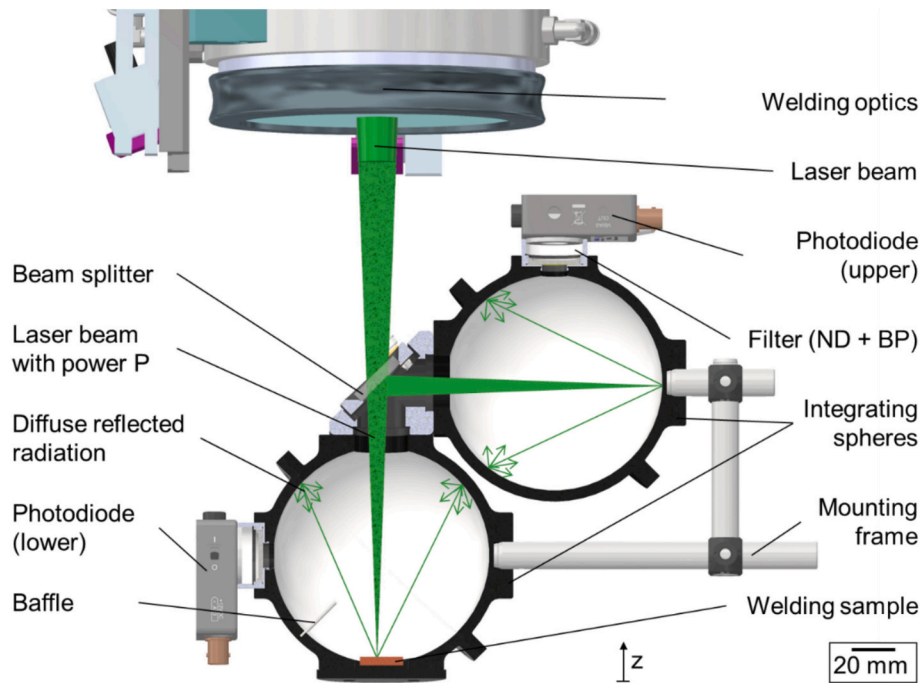


Fig. 3. Double integrating sphere setup for the measurement of the reflected radiation during laser beam welding.

3.3. Setup calibration and data evaluation

By the calibration of the integrating sphere setup, the absolute absorbed laser power can be calculated from the measured photodiode voltage signals. The procedure described below was used to recalibrate the experimental setup for each series of measurements. Exemplary data are shown here for $\lambda = 515$ nm laser radiation.

For the lower integrating sphere, a highly reflective mirror (broad-band mirror Thorlabs BB1-E02 for $\lambda = 515$ nm with a reflectivity of 99.22 %, BB1-E03 for $\lambda = 1030$ nm with a reflectivity of 99.75 %) was installed in the sample port using a mirror mount with an inclination angle of 10° . This angle ensured that laser radiation reaching the mirror was reflected into the lower integrating sphere and not vertically back

out of it. A direct irradiation of the lower photodiode was prevented by the baffle. To calibrate the upper integrating sphere, the mirror was positioned horizontally in the mount. Thus, the incident power was re-reflected coaxially, and the portion reflected at the beam splitter reached the upper integrating sphere, where the signal of its diffuse reflection at the upper photodiode was measured. During this procedure, the laser beam was defocused from the mirror surface in order to avoid any damage of the components.

The laser power was successively increased, and the resulting diode voltages in both mirror positions were recorded. The diode voltages $U_{L,T}$ and $U_{U,T}$ (portion of laser power reflected into the upper sphere during calibration of the lower integrating sphere) correspond to the calibration measurements with tilted mirror, and $U_{U,H}$ is the upper diode voltage

with horizontally tilted mirror. The calibration graphs for all three calibration factors are displayed in Fig. 4 showing good linear trend behavior of the experimental setup.

Using the measured diode voltages U_L and U_U during the welding process, the absorptivity was calculated according to Eq. (3)

$$A = \left(1 - \frac{U_L}{U_{L,T}} - \frac{U_U - U_L \cdot \frac{U_{U,T}}{U_{L,T}}}{U_{U,H}} \right) * 100\%, \quad (3)$$

The first subtraction term denotes the radiation reflected by the sample that was measured in the lower integrating sphere, without taking the vertically reflected portion into account. The second term refers to the portion of radiation measured in the upper integrating sphere (share of vertically reflected radiation including correction term for the lower sphere). The average absorptivity, or coupling efficiency for spot and cw-welds was calculated from Eq. (4)

$$\eta_a = \frac{E_a}{E_0}, \quad (4)$$

with the energy absorbed during the welding process E_a , which was derived from the time integral of the absorbed power, and E_0 the input energy based on the feedback from the laser system. The initial signal peak during processing associated with the laser ramp up in combination with the reflection at the sample surface, as also observed in studies in [39], and the drop of the signal corresponding to the emission cut-off, are temporary conditions and not representative of the rest of the welding processes. Therefore, the first and last 500 μ s (5 % of the shortest process investigated) of the measurement data were excluded from this evaluation. An exemplary absorptivity analysis for a cw weld using 1030 nm laser radiation is shown in Fig. 5, indicating the sections to be trimmed both at the beginning and at the end of the welding process.

After laser processing, the weld seams were documented by microscopic images in top view at 5 to 20 $\times$ magnification. Cross-sectional analyses were then carried out in the weld seam symmetry plane or transverse to the welding direction (four equidistant cutting planes; spot weld: one section per experiment). The cut and embedded samples were ground with up to 1000 grit and subsequently polished using a final grain size of 0.25 μ m. After etching (etchant: Fe(III)-Cl), the weld seam dimensions were analyzed, and the morphology of the weld bead was characterized.

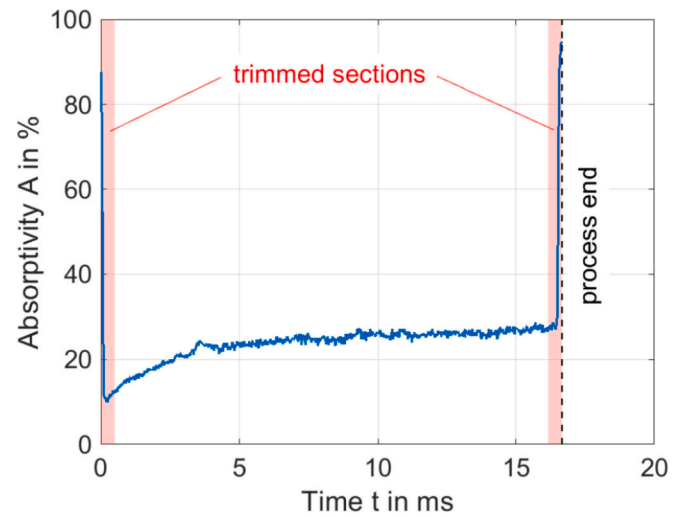


Fig. 5. Dynamic absorptivity data for a cw-weld using 1030 nm laser radiation ($P = 3500$ W, $v = 300$ mm/s) with illustration of the trimmed sections.

4. Results and discussion

4.1. Absorptivity analysis of solid material and HCW-mode

To determine the absorptivity of the material in the solid state when heated with infrared and green laser radiation with identical laser spot dimensions, the lowest possible energy amount (turn-on threshold of the

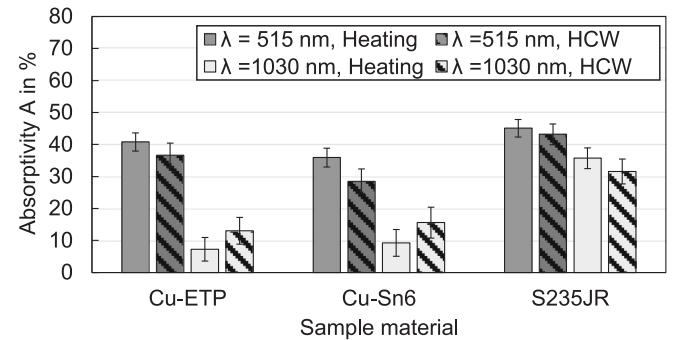


Fig. 6. Measured absorptivity values for heated material and heat conduction welding for different metals for 515 nm and 1030 nm laser radiation using a double integrating sphere setup.

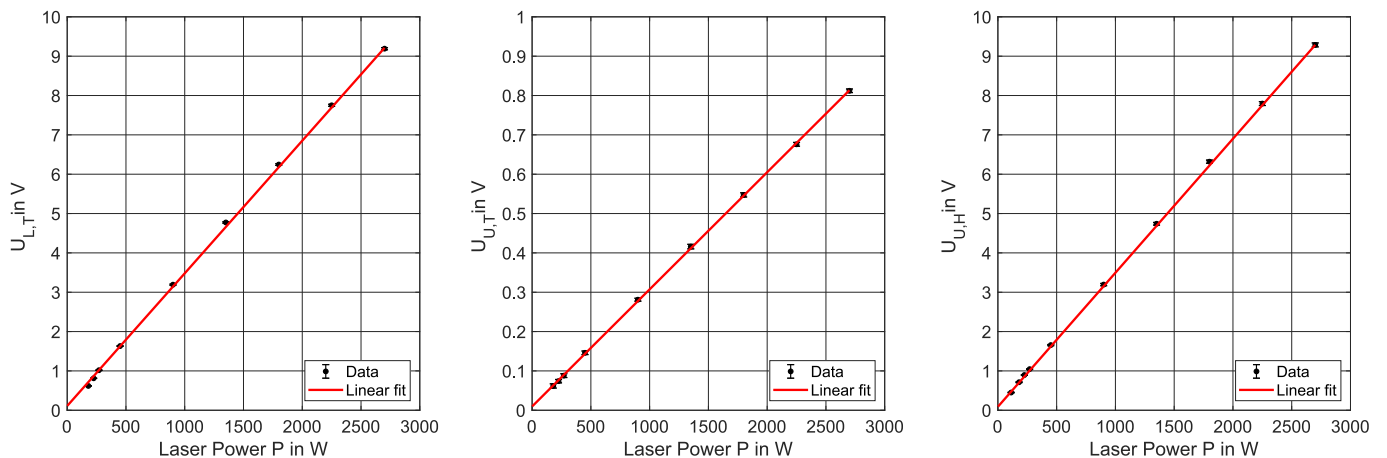


Fig. 4. Calibration curves and linear fits for calibration factors of the integrating sphere setup for laser beam welding with green laser radiation.

laser systems) was irradiated using a defocused beam. The calculated results are shown in Fig. 6 in combination with measurements determining the property in the melt (heat conduction welding, hatched bars). The error bars represent the statistical standard deviation from a set of $n = 4$ individual measurements.

The values and trends for the copper materials (Cu-ETP, CuSn6) together with those for mild steel are in good agreement with literature data as for example [40], also see Fig. 1. These findings demonstrate the functionality of the designed experimental setup and the calibration procedure for data evaluation. Throughout the experimental campaign, no significant change was observed in the calibration factors of the individual test series. It can therefore be assumed that contamination of the BaSO₄ surface has been effectively counteracted, proving the suitability of the test rig as a cost-effective method for the in-depth investigation of laser welding processes.

For the copper materials (Cu-ETP: Cu > 99.9 %; CuSn6: Cu > 91.8 %), absorptivity values >35 % are measured for the first stage for 515 nm, and single-digit values are found for 1030 nm (Cu-ETP: 7.2 %; CuSn6: 9.2 %). In the HCW regime, a decrease in absorptivity for green laser radiation can be noted, while values rise for infrared laser radiation, both of which are accompanied by an increasing standard deviation, an effect assumed to be related to melt pool movements. For mild steel, data (heating: 45.0 % for 515 nm, 35.7 % for 1030 nm) with a slight negative trend in absorptivity with the change of the aggregate state can be observed for both laser wavelengths. The difference in absorptivity values measured when transitioning to the liquid state compared to results obtained by Engler et al. [23] is assumed to be mainly caused by oxidation during melting, as no gas shielding of the welding zone close to the interaction was used (See also temporal effects of oxidation discussed in Sec. 4.4).

Considering the results for copper, the difference in the change with temperature ($T_M(\text{Cu}) = 1084^\circ\text{C}$) can be attributed to the relative position of the processing wavelengths to an inherent optical property, the X-point of the metal. Its material-specific wavelength (which is

suspected to be linked to the melting temperature of the metal) is characterized by a temperature-independent absorptivity, as the sign of the temperature dependence of the latter alters there from negative to positive for increasing wavelengths [41]. Experimentally measurements reveal values between 560 nm and 575 nm for copper [42]. For steel, however, the wavelength of the X-point was observed within the infrared range (Fe: 1600 nm [43]), reflecting the consistent decreasing behavior with temperature increase for S235JR at the selected processing wavelengths.

For the heating and heat-conduction welding process of copper materials with green laser radiation, the absorptivity values indicate that less laser energy is necessary to heat the materials to liquid state compared to using an NIR wavelength (above the X-point). The further temperature increase (heating up to the vaporization temperature) requires relatively more energy (as the absorptivity A decreases with rising temperature) using green laser radiation, which is understood to be a reason for the wider process window [44]. In general, the absorptivity values of the data studied for copper at 515 nm radiation are comparable to those for mild steel.

4.2. Investigation of pulsed laser processing

In Fig. 7 and Fig. 8 dynamic absorptivity results calculated using Eq. (3) are presented for different input power values during 10 ms irradiation for both laser wavelengths on Cu-ETP. The corresponding microscopic top views of the weld results and cross-sections are shown alongside. The top view taken at a pulse power of $P = 200$ W using green laser radiation shows signs of discoloration due to heating, with no significant microstructural changes evident in the cross-section. At low pulse powers and energies ($P = 8$ J for 515 nm, 20 J for 1030 nm respectively) a spot weld with low aspect ratio was produced that is indicative of heat conduction welding. Higher energies (10 J/ 17.5 J for 515 nm, 30 J/ 35 J for 1030 nm) form a joint of high aspect ratio, indicating that the formation of a vapor capillary was achieved.

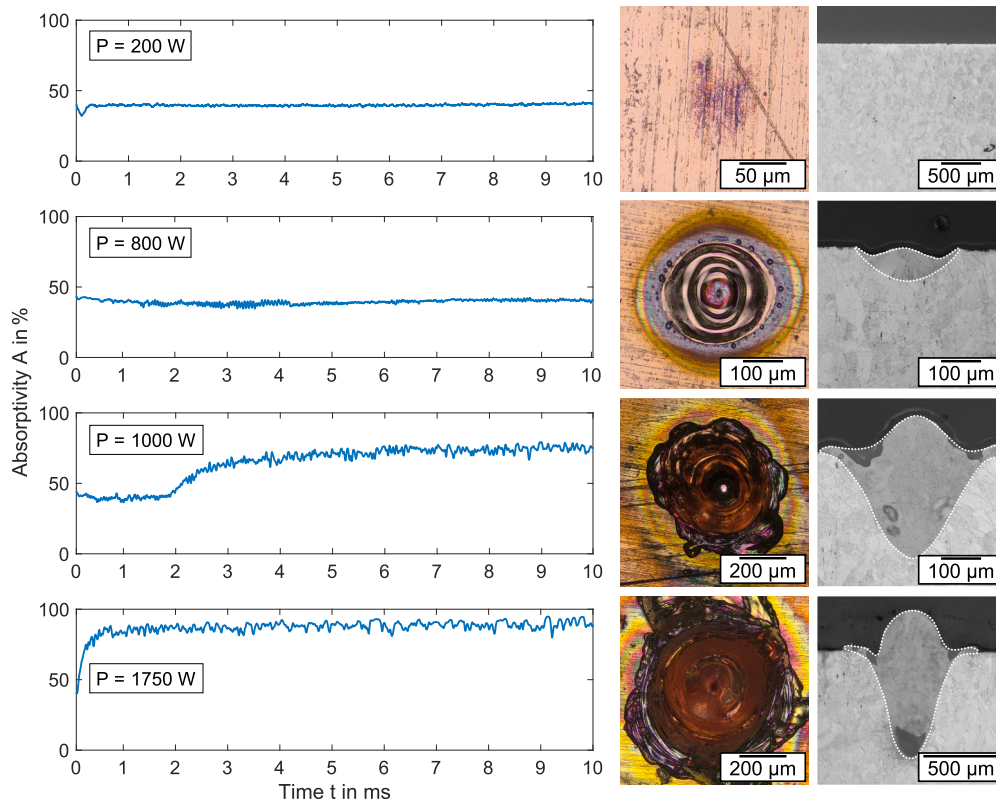


Fig. 7. Measured absorptivity data, top view, and cross-sections of spot welds on Cu-ETP using $\lambda = 515$ nm laser radiation with different pulse powers.

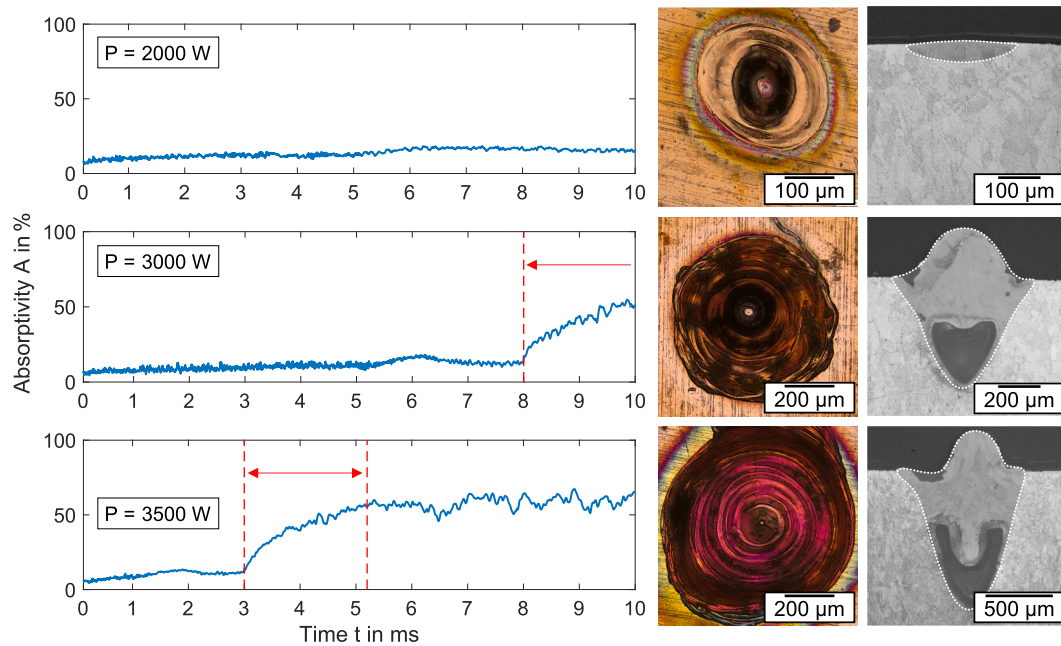


Fig. 8. Measured absorptivity data, top view, and cross-sections of spot welds on Cu-ETP using $\lambda = 1030$ nm laser radiation with different pulse powers.

During heat conduction welding, a higher value of absorptivity is observed for green laser radiation compared to NIR radiation, which is also reflected in a lower aspect ratio of the weld seam. A variation in the graphs at the phase transition (discussed in Section 4.1) can be detected between 4 and 5 ms for 515 nm ($P = 800$ W) and between 5 and 6 ms for 1030 nm ($P = 2000$ W) laser radiation. The more the pulse energy increases, the less pronounced this effect becomes visible in the graphs. At higher input energies, these absorptivity levels are followed by a second rise, which is most pronounced at the highest pulse powers (nearing 85 % absorptivity for 515 nm; 60 % for 1030 nm). As the cross-sections for these welding parameters are characterized by a significantly increased aspect ratio of the weld nugget, the deep penetration welding regime was achieved. This absorptivity increase is therefore identified as the sign of vapor capillary formation, representing the erratic increase in absorption of the irradiated energy due to multiple reflections within the cavity. Increased dynamics in this regime can be recognized from the variation in dynamic absorptivity values, visible in the top views in the form of molten material ejected to the outer edge of the interaction zone and solidified material showing seam overcut. These findings are consistent with experimental work investigating the absorptivity of metals as a function of temperature in [15,31].

With the formation of the vapor capillary, the width of the spot weld suddenly increases to ~ 2 times the focal diameter, particularly noticeable for NIR laser radiation. It is interesting to note that the duration of capillary formation is dependent on the wavelength used: In the case of green laser radiation, no >1 ms elapses between the lower absorptivity level and a sharp rise to a stabilization at an absorption ratio of ~ 70 % for $P = 1000$ W. The use of higher laser power shortens this process, accompanied by a further increase in absorptivity to ~ 85 %. Using infrared laser radiation, the increase before leveling at a pulse power of 3000 W takes at least 2 ms until the end of the pulse (transition point when the capillary formation begins). At 3500 W, the duration is determined to be ~ 2.2 ms (red arrows in Fig. 8), which is considerably longer with a lower absorptivity gradient compared to the results for green laser radiation. This phenomenon is assumed to be related to the effect of multiple reflections, which are fewer in an initially more shallow capillary. As the penetration depth of the capillary increases, their influence is stronger for infrared laser radiation due to the lower absorptivity for copper [45].

The above interpretation is also supported by the investigation of the

average coupling efficiency η_A , computed using Eq. (6). The values over a range of pulse powers for Cu-ETP and CuSn6 for green and infrared laser radiation are shown in Fig. 9. The different welding modes are indicated by the background colour as a guide for the eye (white = HCW; grey = DPW). The error bars indicate the statistical standard deviation from a set of $n = 3$ recordings. The behavior of a step function from a low coupling efficiency with values around the absorptivity in molten state to a transition eventually approaching η_A -values close to 90 % for green laser radiation and lower values for 1030 nm wavelength is clearly visible for Cu-ETP and CuSn6. Higher pulse powers were not investigated due to increased spatter formation, impacting the sensible surface coating of the experimental setup.

In general, the coupling efficiency trends for Cu-ETP and CuSn6 are comparable, however, the threshold values for deep penetration welding for CuSn6 are lower compared to pure copper. This is related to the optical properties and the thermal conductivity of the metals used, as the alloying deteriorates this physical property, causing increased heat accumulation in the interaction zone. Increased standard deviation is noted for pulsed laser processing with 1030 nm especially in the transition area which can be mainly ascribed to the HCW share of the dynamic absorptivity for these parameter sets. The higher the pulse power, the weaker this contribution, reducing the standard deviation of the data points, accompanied by a higher reproducibility of the results. The overall phenomenon has also been described in studies for other metals, for example for aluminum and stainless steel [46]. From the results, a more efficient and stabilized processing using green laser radiation can be concluded, which is in agreement with investigations on pulse size stability and the observation of incoupling behavior for both laser wavelengths presented in [47]. The detected pulse powers for Cu-ETP for the first observation of deep penetration welding of about 900 W for 515 nm and < 3000 W for 1030 nm, related to the spot diameter, are in the same order of magnitude as discussed in [48]. Thus, the improved absorptivity in laser-matter interaction, both in solid and liquid state, reduces the energy required to reach the melting and vaporization temperature of the copper materials with 515 nm laser radiation, broadening the process window for deep penetration welding.

4.3. Influence of surface condition

To evaluate the influence of different surface conditions on the

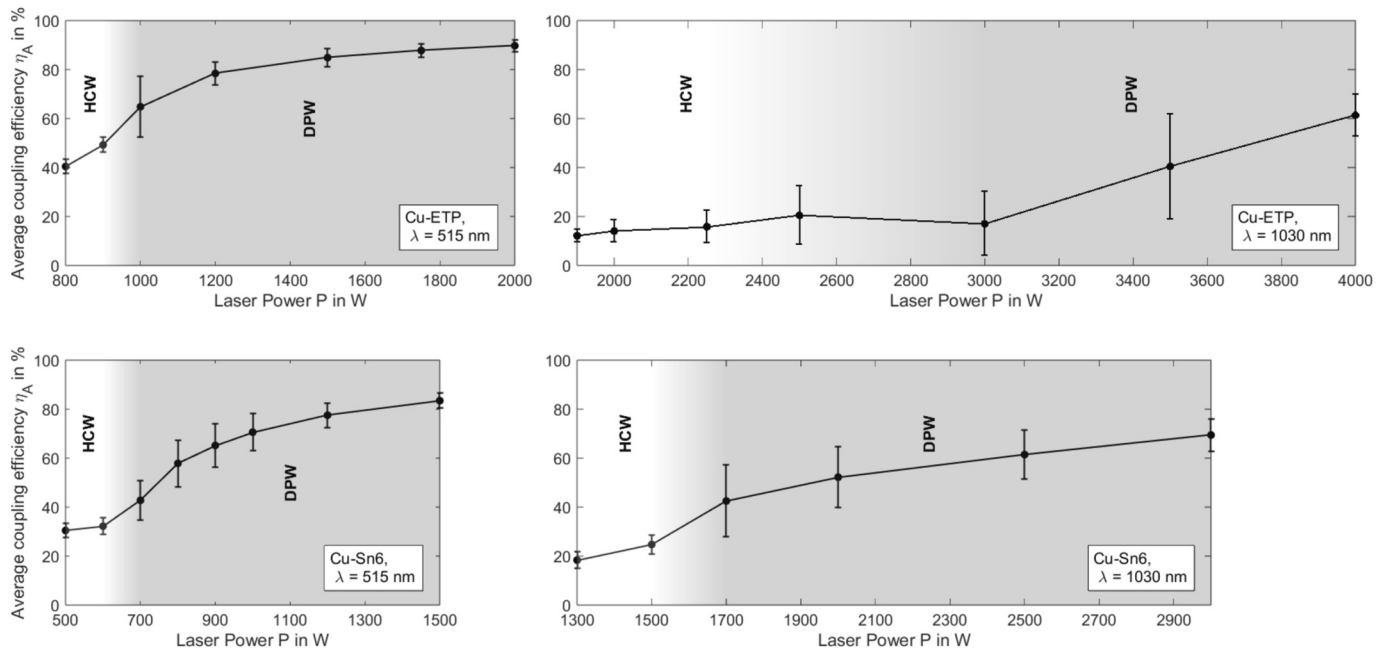


Fig. 9. Average coupling efficiencies as a function of laser power during a 10 ms spot weld (pulse) for Cu-ETP and CuSn6 using green and infrared laser radiation.

energy coupling, measurements conducted on Cu-ETP samples prepared by sandblasting are compared to the results for untreated material (only cleaned, see Sec. 3.2) discussed in the previous sections. For laser powers for which both HCW and DPW shares in the dynamic absorptivity data were identified, measurement results for processing with different surface conditions are displayed in Fig. 10 using 515 nm and 1030 nm laser radiation.

By surface roughening, the reflectivity of Cu-ETP can be reduced at room temperature in a wavelength range from 400 to 1100 nm, as has been demonstrated in [14]. This effect leads to increased absorptivity for the roughened samples (see red arrows and sketch illustrating the phenomenon) at the beginning of the laser irradiation, visible for both laser wavelengths in the orange colored curves. With progressing heating, the surface structures get flattened when approaching the melting temperature of copper (formation of a mirror-like surface at the beginning of heat conduction welding), causing the absorptivity values to decrease. Here, comparable absorptivity values with those recorded before vapor capillary initiation on the untreated samples arise, see red dotted lines as a guide for the eye. This reduction is less pronounced for green laser radiation, as the capillary formation immediately follows. An earlier

transition to deep penetration welding mode is also recognizable for infrared laser radiation with the pre-treated sample compared to an untreated surface. This behavior is attributed to the increased energy input into the metal caused by the observed enhanced initial absorptivity in the case of roughened sample surfaces. When calculating the coupling efficiency for the parameter sets, an η_A -increase of 5 % is found for 515 nm laser radiation, and a gain of 117 % between the different surface states becomes apparent for 1030 nm laser radiation. This suggests that roughening is of minor importance when welding copper with visible laser radiation. At higher pulse powers and energies respectively, this phenomenon is less visible in the recorded absorptivity data. In the case of continuous line welds (cw-laser processing), no significant difference in the weld results and absorptivity data was observed for green and infrared wavelengths for different surface conditions, apart from the effect discussed here for pulsed processing. However, the initiation of a DPW process can vary depending on the process parameters and surface preparation. The detailed analysis of cw-laser processing of Cu-ETP is presented in the following section.

4.4. Investigation of cw-laser processing

To quantify important characteristics including the points of melting and vapor capillary formation when welding copper materials with infrared and green laser radiation and moving laser beam, Fig. 11, and Fig. 12 illustrate measured absorptivity plots for different feed rates between $v = 100\text{--}300\text{ mm/s}$ with continuously increased laser power P . To the right, the corresponding microscopic images of the weld results are given.

Starting from the lowest laser power used, the material is heated from the solid state to the melting temperature, associated with a reduction in absorptivity for 515 nm (laser power range 500 to 550 W), as previously observed in Sec. 4.1 and 4.2. In the microscopic images, the formation of the melt pool can then be recognized by the continuous increase in weld seam width with the forward movement of the beam. A further increase in laser power, however, does not yet lead to a significant change in absorptivity. The waviness in the curves (absorptivity increase from 550 to 600 W with oxide formation, then decrease with melt pool enlargement until capillary initiation) is attributed to weak effects of oxidation, similar to the observations in [49], which can be

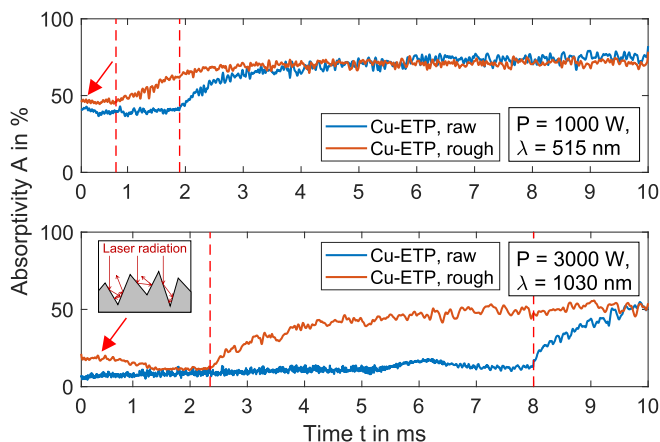


Fig. 10. Measured absorptivity data of 10 ms spot welds on Cu-ETP using $\lambda = 515\text{ nm}$ and 1030 nm laser radiation with different sample surface conditions.

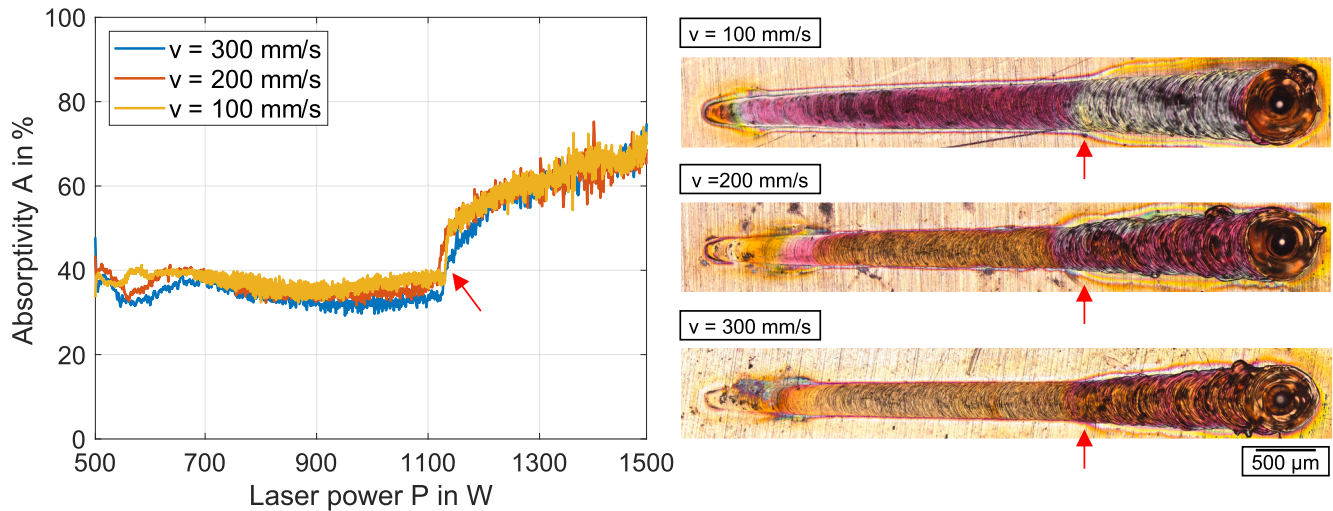


Fig. 11. Measured absorptivity for Cu-ETP for power ramps from 500 to 1500 W using a wavelength of 515 nm; microscopic top views of the seams to the right with the deep penetration threshold denoted by red arrows. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

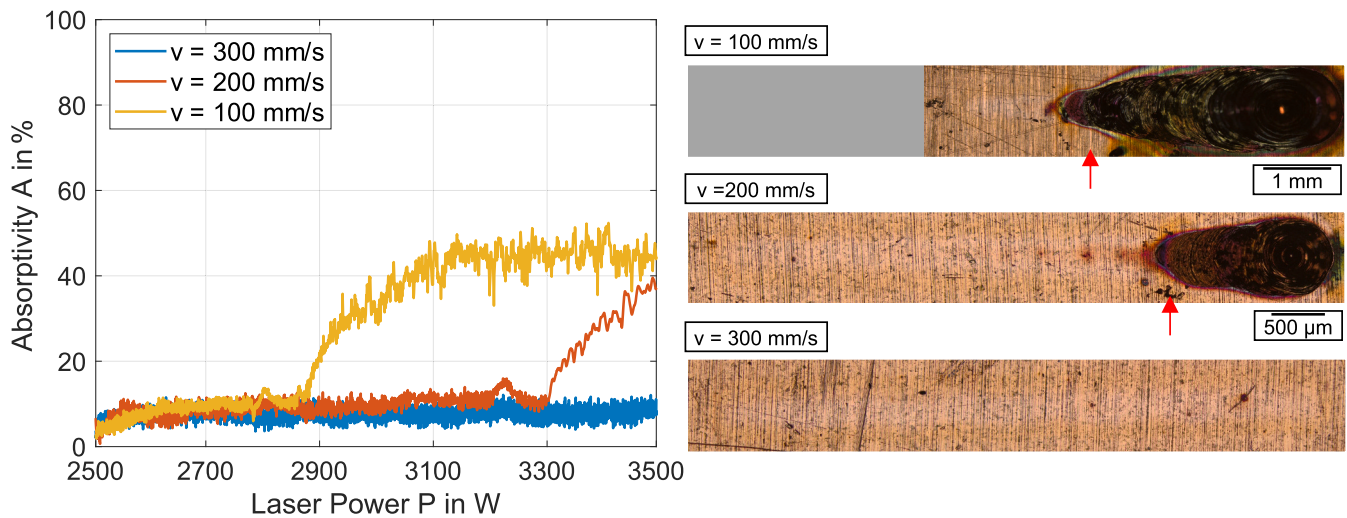


Fig. 12. Measured absorptivity for Cu-ETP for power ramps from 2500 to 3500 W using a wavelength of 1030 nm; microscopic top views of the seams to the right with the deep penetration threshold denoted by red arrows.

recognized in the colour change of the seam surface and in its surrounding area. From ~ 1000 W, the formation of a vapor capillary starts, evident from an abrupt widening of the weld bead (see red arrows beneath the weld results) and the increase in measured absorptivity (red arrows in the diagram) from the integrating sphere setup. This width increase in DPW mode is lower when the feed rate increases, as to influence of heat dissipation decreases when the welding process is accelerated [50]. At 515 nm, almost no dependency on the feed rate is observed for the deep penetration threshold, and the variation of absorptivity data between different feed rates is low.

For 1030 nm laser radiation, in contrast, such clear trends and transitions are not detectable. Single digit absorptivity values at the beginning of the weld indicate solid material and moderate heating, as most of the irradiated energy is reflected in the experimental setup. A change in colour for the laser trajectory can be recognized on the sample surface. The identification of a HCW regime is hardly feasible, only slight signs of further heating before the direct formation of the capillary (DPW) are apparent for the feed rates of $v = 100$ mm/s and 200 mm/s. These are visible in a tooth-like pattern of the absorptivity before the capillary is initiated. The absence of a stable heat conduction welding is

in good accordance with literature reporting that this mode is hardly controllable. The increase in absorptivity values becomes evident with deep penetration welding, observing maximum values of $>40\%$ for the highest laser power. At the highest feed speed of $v = 300$ mm/s, the initiation of a welding process could not be observed. This feed rate dependency of the welding regimes for infrared compared to green laser radiation is in good agreement with the results obtained in [48]. In general, the deep penetration thresholds for copper material are found increased compared to the values determined for pulsed laser welding, mainly due to the movement of the laser spot and the therefore increased influence of heat dissipation into the sample material (stationary processing compared to moving laser beam). Comparable characteristics of the melt pool dynamics in terms of fluctuating measurement results can be derived from the time-dependent absorptivity. Results for constant laser power in the deep penetration regime are shown in Fig. 13 for green and Fig. 14 for infrared laser radiation for Cu-ETP.

Here the dynamic absorptivity for two laser powers levels with a distance of 500 W for characteristic process speeds of $v = 100$ – 300 mm/s is summarized. Since the deep penetration threshold for 1030 nm for Cu-ETP is higher according to the investigations presented, increased

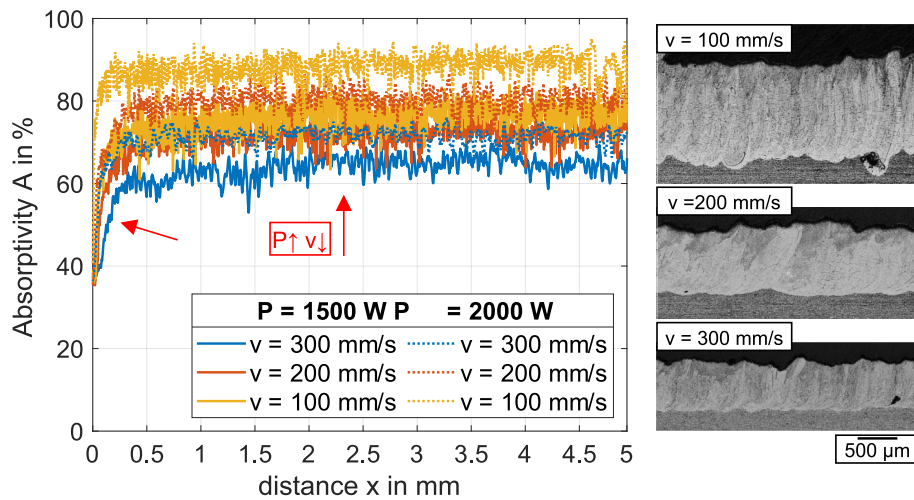


Fig. 13. Measured absorptivity for Cu-ETP for laser powers of $P = 1500$ W and 2000 W and feed rates from $v = 100$ – 300 mm/s using 515 nm laser radiation; longitudinal weld seam cross-sections for $P = 2000$ W to the right. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

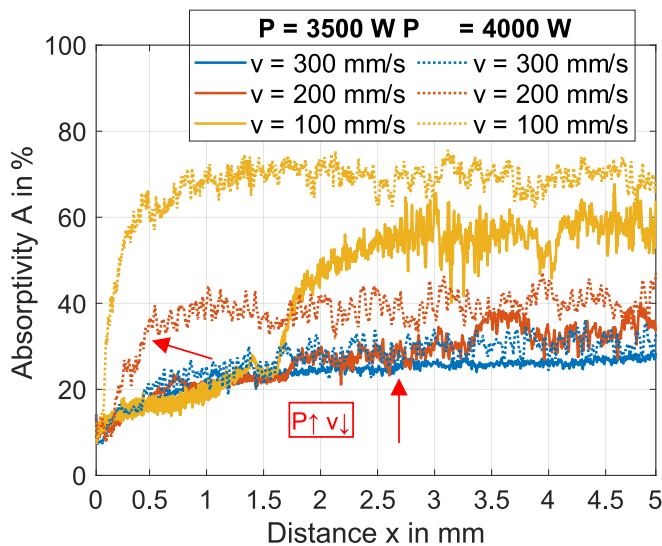


Fig. 14. Measured absorptivity for Cu-ETP for laser powers of $P = 3500$ W and 4000 W and feed rates from $v = 100$ – 300 mm/s using 1030 nm laser radiation.

laser powers compared to green radiation were used for this purpose. The x-axis of the diagrams represents the distance from the start of the weld seam in mm, as the feed rate variation leads to adapted welding times ranging from 16.67 ms to 50 ms. From the data it can be concluded that with increased feed rate the absorptivity (and η_A) decreases, as the capillary is less deep, multiple reflections decrease with this, and less energy is deposited in the processed material. Increasing the laser power counteracts this trend, because this increase raises the penetration depth and causes a less inclined capillary when the feed rates is the same [51]. The observed penetration depth change for various parameter sets can thus be evaluated using quantitative absorptivity values for the processing of copper materials, which are indicated for both wavelengths investigated. With 1030 nm laser radiation, there are still heat conduction sections noticeable in the data due to the high reflectivity of the material for this wavelength, particularly at increased feed rates. Nevertheless, the general trends described are evident for infrared and green laser radiation, illustrated by the red arrows and boxes in the graphs. However, the values differ depending on the wavelength used, so absorptivities close to 90% are found for 2 kW laser power for green

laser radiation, whereas the values for $P = 4$ kW infrared radiation with the same beam diameter were only measured at $\sim 70\%$. For the dynamic absorptivity behavior at the beginning of the welding process, an accelerated rise in the values towards a quasi-steady state can be detected with increased laser power and reduced feed rate. With increased processing speed, a gradual decrease in the dynamic range of the signal is also noticeable. This effect is assumed to be related to the observed increased process stability with higher feed rate, due to an increased capillary front wall angle in combination with enhanced evaporation dynamics [45]. In the longitudinal cross-sections of the weld seam using 515 nm laser radiation, the needle like appearance of the lower seam edge for $v = 100$ mm/s is observed to change to a more uniform appearance for $v = 200$ and 300 mm/s.

The experimental findings discussed in this work are consistent with studies on the absorptivity of copper as a function of welding parameters in [36]. The conclusions of Wittemer [34] stating that the process stability can be assessed using the integrating sphere signals, as there is a correlation between signal volatility and the stability of the weld seam characteristics, can also be confirmed with our results. Fig. 15

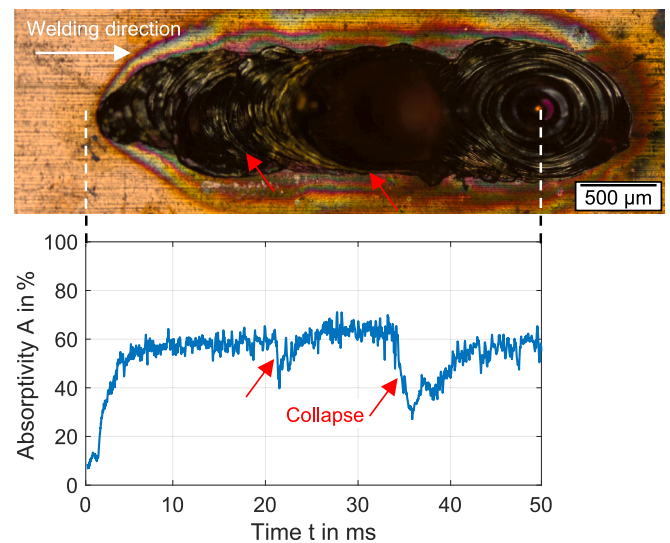


Fig. 15. Vapor capillary bulging and collapse with melt ejection in microscopic top view and corresponding absorptivity plot (bottom) for a weld on Cu-ETP with $P = 3750$ W, $v = 100$ mm/s using a wavelength of 1030 nm.

additionally shows a characteristic absorptivity graph of a copper weld using a wavelength of 1030 nm with striking zones and the corresponding microscopic image above.

It can be noted that the absorptivity progression of the weld with $P = 3750$ W at a feed rate of $v = 100$ mm/s follows the typical behavior determined and discussed above. The absorptivity is found to stabilize at a level of $\sim 60\%$, where significant dynamic behavior can be detected. At around 21 ms, the absorptivity curve abruptly decreases by approx. 20 % (red arrow as a guide to the eye). However, with slightly higher values compared to the previous level, the welding process seems to stabilize again quickly. In the top view, a disturbance in the surface of the solidified material in this area can be seen. This phenomenon is thought to correlate with the capillary instabilities observed in the in situ X-Ray analysis in [52] at slow feed rates on Cu-ETP. With the bulging of the capillary to its rear end, the energy coupling varies dynamically. An additional amount of energy may be absorbed in the sample due to increased multiple reflections in the capillary (cavity area), reflected in the increased absorptivity values measured after the first major instability. The resulting increase in evaporation pressure has to be compensated for by the acting closing forces of the capillary. If this pressure balance is disturbed, parts of the molten metal, and in extreme cases the entire melt pool, can be ejected from the interaction zone. The crater-shaped region at the top of the seam, also reported in the literature, indicates the occurrence of this effect. This state is accompanied by a significant loss of absorptivity, since the aperture at the sample surface is greatly increased during this period, causing high losses due to back-reflected radiation. As most of the capillary is subsequently reformed, it takes considerable time for the measured absorptivity curve to return from 30 % to values comparable to the average before the end of the welding process.

Investigations presented in [53] also observe and correlate a change in the back reflection signal with changed capillary shape during welding of AISI 304. For the experiments using CuSn6, similar absorptivity trends are observed in this work, however the absolute values are different. In particular, the deep penetration threshold for 1030 nm radiation is found to be less sensitive to feed rate variations and, as analyzed for the pulsed welding process, is reduced compared to pure copper. In the following section, in conjunction with analytical models, the general behavior of the coupling efficiency for both materials (Cu-ETP and CuSn6) is further analyzed.

4.5. Influence of the capillary shape on the coupling efficiency

For a comparison of the results of continuous line welds on copper materials using green and infrared laser radiation, the calculated coupling efficiency is plotted against the aspect ratio of the weld seam in Fig. 16. Using this ratio, the shape of the capillary can be characterized.

Various laser powers and feed rates from 100 to 300 mm/s are taken into account here. Additionally, the results for mild steel are presented on the right.

Considering the model of multiple reflections of rays on a black body proposed by Gouffé [54] together with the refinements presented by the authors in [55], the coupling efficiency can be evaluated analytically. Based on the assumption of a cylindrically shaped vapor capillary in liquid copper, and with a mean absorptivity calculated from the solid and liquid state values for the corresponding material and wavelength, η_G can be calculated according to Eq. (5)

$$\eta_G = A \frac{1 + (1 - A)(s - \beta)}{A(1 - s) + s}, \quad (5)$$

with a calculated model aspect ratio s

$$s = \frac{1}{2 \left(1 + \frac{e}{d_F} \right)}, \quad (6)$$

and the angle β

$$\beta = 4\pi \sin^2 \left(\frac{\tan^{-1} \left(\frac{d_F}{2e} \right)}{2} \right), \quad (7)$$

using the geometric characteristics welding depth e and diameter d_F . The results of the calculation are added as lines in Fig. 16. Please note that the model results were transferred to the x-axis of the measured aspect ratio e/d_F .

For all material and wavelength combinations, the average coupling efficiency is observed to rise as the aspect ratio e/d_F of the weld increases. This phenomenon can be targeted by a suitable choice of the process parameters laser power and feed rate, as the capillary shape is significantly influenced by these (see red arrow indicating the trend in the CuSn6 plot, which applies for all materials investigated). A significantly lower coupling efficiency in case of welding with a small aspect ratio becomes visible, especially for pure copper when using infrared laser radiation. When comparing capillaries with constant aspect ratio, an average increase in coupling efficiency of 60 % is observed for Cu-ETP applying 515 nm laser welding compared to 1030 nm laser radiation. This difference can be attributed to the optical properties of the material, but also partly to phenomena described in the previous section, negatively affecting the efficiency. The wavelength-dependent difference in energy coupling is observed to be less pronounced for the CuSn6 bronze and the mild steel S235JR. Here the green laser radiation shows only slightly higher degrees of energy coupling at comparable aspect ratios of 17 % on average for CuSn6 and 6 % for steel. The

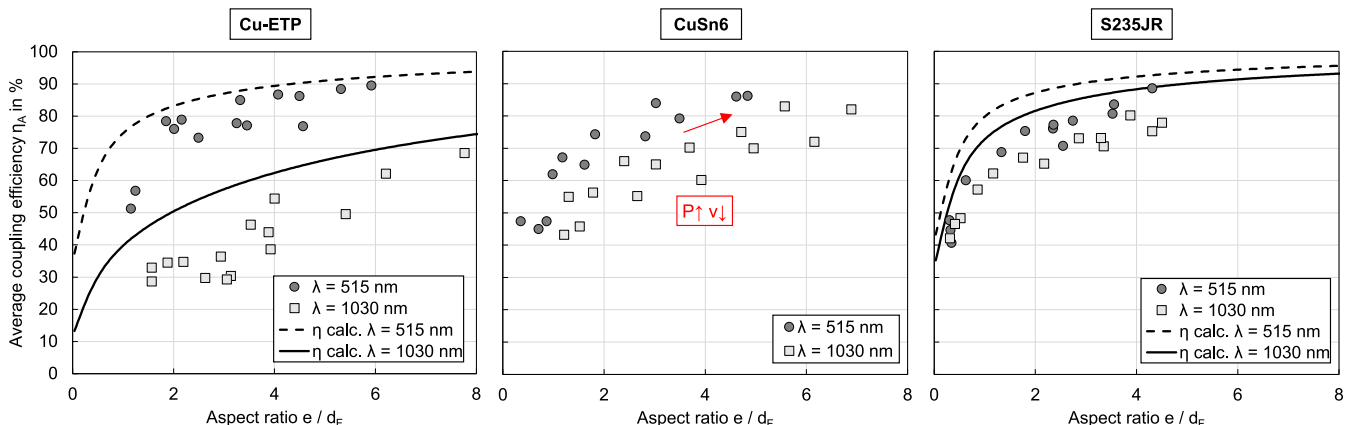


Fig. 16. Coupling efficiency of capillaries in dependence on their aspect ratio for different metals for infrared and green laser radiation.

quantitative results for steel of about 60–90 % are also in good agreement with the experimental results reported in [56]. Additionally, recent studies for other materials using an infrared laser show a similar increase of the absorptivity [57]. The geometrical model of Gouffe generally gives higher values than those measured experimentally owing to the omission of hydrodynamic influences on the capillary and the melt pool behavior, which are decisive for the process dynamics. This effect is more pronounced in the case of copper materials, where the formation of spatter is known to be pronounced [58]. The results show that by using the deep penetration welding process on Cu-ETP, average coupling degrees of >80 % can be achieved on Cu-ETP by applying visible laser radiation. Overall, the differences between the two wavelengths become smaller in this regime due to the effect of multiple reflections compared to the absorptivity investigations in the solid state and HCW regime of the materials. For higher aspect ratios of the capillary, this difference decreases further as this factor gains importance.

5. Conclusion

Current theoretical models do not adequately reflect the phenomena of energy coupling under real process conditions in laser welding of metals. Moreover, knowledge of the influence of temperature and related process parameters on absorptivity is limited but essential for an efficient process design. This is of particular interest in the laser based processing of copper materials, which are widely used these days due to electrification and digitalization efforts. Therefore, comprehensive experimental analysis of energy coupling effects under real copper processing conditions is essential. In this work, the investigation of the absorptivity and coupling efficiency was performed using a specially developed, low-cost experimental setup based on integrating spheres. The material and wavelength specific process behavior was investigated by performing welding trials with different laser wavelengths of $\lambda = 515$ nm and $\lambda = 1030$ nm with spot diameters of comparable value on different substrates. The measurements, taken with microsecond resolution over a wide range of processing parameters, were also correlated with metallographic analysis of the welds in order to expand the knowledge of the underlying effects. The following conclusions can be drawn:

- Significantly increased absorptivity values for copper materials are observed during solid heating and heat conduction welding using green laser radiation, requiring less energy input to reach the melting temperature, and obtaining a higher aspect ratio of the melt lenses in the first welding regime compared to infrared laser radiation.
- The evaluation of melting and vapor capillary formation during pulsed laser processing determined from the dynamic absorptivity analysis showed a significantly increased temporal variation in the results for 1030 nm laser radiation in combination with a slower increase in absorptivity at the capillary onset. This behavior is attributed to the increased influence of multiple reflections for infrared radiation.
- Considering the influence of the surface condition, roughening of the sample surface is of major importance when welding copper with infrared laser radiation. At the beginning of the sample irradiation, a significant increase in the absorptivity curves was recognized for each of the two laser wavelengths, which decreased as the irradiation progressed. The increased energy coupling accelerates capillary formation, especially in the case of 1030 nm laser radiation.
- Regarding the onset of the deep penetration welding mode, the green laser radiation is significantly less sensitive to the feed rate compared to infrared laser radiation due to the increased initial absorptivity. With the formation of the capillary, the absorptivity rises significantly and is measured at values of up to 90 %.
- Signal fluctuations are found to vary depending on the parameter selection. Erratic changes in the measured absorptivity during the

welding process could be correlated with the dynamic behavior of the vapor capillary, characteristic of an unstable and inefficient welding process.

- The deep penetration welding process leads to an efficient use of the irradiated energy, with our results confirming the scaling of the average coupling efficiency with the aspect ratio of the capillary. Measurement data reveals that the difference in energy coupling is less significant in this regime for 515 nm compared to 1030 nm laser radiation when processing copper materials, but is particularly beneficial for low aspect ratio capillaries.

Consent to participate

Not applicable.

CRediT authorship contribution statement

Florian Kaufmann: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Robert Möttingdörfer:** Methodology, Investigation. **Stephan Roth:** Writing – review & editing, Supervision. **Michael Schmidt:** Supervision, Project administration, Funding acquisition.

Ethics approval

The authors respect the ethical guidelines of the journal.

Consent for publication

Not applicable.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data sets generated and analyzed during the current study are available from the corresponding author on reasonable request.

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